



ZEAL EDUCATION SOCIETY'S
ZEAL COLLEGE OF ENGINEERING AND RESEARCH
NARHE | PUNE -41 | INDIA



Record No.: ZCOER-ACAD/R/16M

Revision: 00

Date: 01/04/2021

Question Bank

Department:

Semester: II

Academic Year: 2024-2025

Class: F.Y.B.Tech.

Div:

Date:

Course: Engineering Mathematics II

Unit No. - name	Q. No.	Question	Marks	CO	Blooms Level
Application of Multiple Integral	1.	Find the area bounded by the curves $y = x^2$ and $y = x$.	05	CO6	2
	2.	Find the area between the curves $y^2 = x$ and $x^2 = -8y$	05	CO6	2
	3.	Find the volume of the cylinder $x^2 + y^2 = 2ax$ intercepted between the paraboloid $x^2 + y^2 = 2az$ and the XY plane .	05	CO6	2
	4.	Prove that volume bounded by cylinder $y^2 = x$, $x^2 = y$ and plane $z = 0, x + y + z = 2$ is $\frac{11}{30}$.	05	CO6	3
	5.	Evaluate $\iiint x^2 y z \, dx \, dy \, dz$ throughout the volume bounded by the plane $x = 0, y = 0, z = 0, \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.	05	CO6	2
	6.	Find the y-coordinate of the center of the area bounded by $r = a \sin \theta$ and $r = 2a \sin \theta$. Given that the area bounded by these curves is $\frac{3\pi a^2}{4}$.	05	CO6	2
	7.	Find the x-coordinator of center of gravity of a loop of $r = a \sin 2\theta$ in first quadrant ,given that area of loop is $A = \frac{\pi a^2}{8}$.,	05	CO6	2
	8.	Find the x-coordinator of center of gravity of a loop of $r = a \cos 2\theta$ in first quadrant ,given that area of loop is $A = \frac{\pi a^2}{8}$.,	05	CO6	2
	9.	Prove that the moment of inertia of the area included	05	CO6	3



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		between the curves $y^2 = 4ax$ and $x^2 = 4ay$ about x axis is $\frac{144}{35} M a^2$, where M is the mass of the area included between the curves. Given that density is $\rho = \frac{3M}{16 a^2}$			
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Course faculty